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Speakers as sensors: Machine learning-based earphone impedance analysis for ear canal insertion depth prediction

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ABSTRACT:
This study investigates the use of speakers (earphones) as sensors for detecting acoustic loads through variations in electrical impedance. Leveraging electroacoustic coupling and machine learning-based analysis, impedance features were used to predict insertion depth with high accuracy. The dense neural network-based classification achieved 87% overall accuracy, improving to 91% with speaker-specific data and up to 99% for individual units. By identifying optimal frequency-domain regions in which impedance responses were most sensitive to changes in acoustic loading, classification performance further improved. The results were also compared with a regression model, which yielded higher error rates but indicated a promising direction for future work. This work integrated electroacoustic principles with machine learning to establish a foundation for smart, low-cost, and portable diagnostic systems for ear health and related biomedical applications. © *2026 Acoustical Society of America*. https://doi.org/10.1121/10.0043165

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I. INTRODUCTION

Microphones and speakers are indispensable components in hearing care and audio research, as they are electroacoustic transducers that convert between electrical and acoustic energy.[1] In hearing-related applications, these devices enable noninvasive diagnostics and monitoring of auditory function (e.g., otoacoustic emission measurements, ear-canal acoustic loading assessment, and *in situ* hearing-aid or earphone characterization) through assessments of cochlear and ear-canal acoustics.[2,3] These same sensor–actuator elements form the foundation of modern earphone-based systems: Modern earphones now integrate both microphones and speakers in a single enclosure to support bidirectional audio interaction, such as noise cancellation and adaptive sound processing.

From a physical perspective, both microphones and speakers are governed by coupled electrical, mechanical, and acoustic domains, where changes in acoustic boundary conditions are reflected in measurable electrical.[4] This reciprocity (or anti-reciprocity for electromagnetic type transducers) principle underlies many impedance sensing approaches and motivates the exploration of electromechanic and/or electroacoustic transducers not only as signal actuator but also as sensing elements.[5]

The impedance-based sensing has been widely explored for structural diagnostics. Electromechanical impedance techniques, for instance, employ piezoelectric sensors to monitor impedance changes that reveal structural defects or material degradation.[6] These methods have proven effective in structural health monitoring but face persistent challenges: sensitivity to temperature, frequency, and bonding conditions, and difficulty in localizing damage accurately.[7–11]

Recently, Kim and co-workers extended such sensing modalities into detecting acoustic load characteristics in the human ear canal by converting conventional speakers to act as both actuators and sensors.[5,12–14] Specifically, the earphones functioned as sensors by measuring changes in their electrical impedance under different acoustic loading conditions, and those impedance responses were subsequently post-processed (image-converted and classified) using artificial intelligence (AI) to infer the acoustic environment. Variations in the acoustic load modify the speaker's electrical impedance through electroacoustic coupling, such as shifts in resonance frequency, magnitude, and phase response, thereby providing an indirect but information-rich sensing modality. However, in such bioacoustic environments, the acoustic load is inherently variable and nonlinear, influenced by geometry, compliance, and boundary conditions. To capture these highly nonlinear relationships, the study repurposed earphones, conventionally used as actuators, as sensors by using electrical impedance responses that changed by acoustic load statuses, such as open or blocked syringe lengths. Building on this, they image-converted impedance data and developed an AI-based classification model capable of inferring controlled acoustic loading conditions.[15–17] This work demonstrated the feasibility of "speaker-as-sensor" technology, but it also displayed limitations in terms of data efficiency and model scalability. These

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limitations inspired the present work, which seeks to improve data utilization and generalizability by refining feature representations, frequency-region selection, and model design.

The motivation of the present work is to establish a more direct, physics-aligned sensing framework that reduces preprocessing complexity while preserving sensitivity to acoustic boundary changes. Building on this foundation, the present study advances the concept by directly utilizing raw electrical impedance signals instead of image-transformed data, enabling faster processing and higher fidelity feature extraction. By operating within the audible frequency range (20 Hz–20 kHz), the proposed approach maintains direct relevance to hearing-device applications while leveraging impedance features that reflect acoustic compliance and volume changes. We engineer the speaker to function simultaneously as both actuator and sensor enabling the collection of acoustic information that is subsequently analyzed using machine learning (ML). This improvement provides a technical foundation for scalable, speaker-based impedance sensing technology that can inform future hearing care monitoring and biomedical diagnostics systems.

The scientific contribution of this work lies in integrating electroacoustic sensing principles with data-driven inference, enabling acoustic boundary conditions to be inferred from electrical measurements in a manner not previously explored in conventional electroacoustic modeling. To analyze the relationship between measured electrical impedance and the corresponding acoustic loading conditions, we adopted a data-driven approach based on supervised ML approach [implemented using a dense neural network (DNN)]. In supervised learning, a computational model is trained using examples in which the input data are paired with known physical labels, allowing the model to learn how changes in the input relate to changes in the system state.[18,19] Advances in deep learning have demonstrated that layered neural network models can effectively approximate complex nonlinear relationships across a wide range of scientific and engineering domains.[20]

In this study, we employed a DNN, a class of ML models composed of interconnected computational layers that progressively learn nonlinear relationships between input features and output variables. Although data-driven methods are still relatively new to much of the acoustics research community, they offer a practical means to analyze electroacoustic systems in which multiple coupled physical mechanisms make closed-form modeling difficult. Here, statistical learning is used as an inference tool to extract physically meaningful patterns embedded in electroacoustic impedance measurements, rather than as a replacement for acoustic or electroacoustic theory.

II. METHODS

A. Experimental setup

The experimental configuration in this study builds upon the framework introduced by Kim *et al.*,[13] in which earbud speakers were previously characterized as coupled actuator–sensor elements by measuring their electrical impedance under controlled acoustic loading conditions. In that prior work, a syringe-based enclosure was used for precise manipulation of acoustic boundary conditions and speaker positioning, enabling systematic variation of the insertion length.

Four distinct earbud speakers, labeled A, B, C, and D, were evaluated (Table I). All devices are passive wired earphones with no active electronics, and the measurements reflect the intrinsic electroacoustic properties of the speaker transducers and their housings, including differences in geometry and mechanical loading. To ensure a consistent acoustic seal and to mimic the coupling between an earbud and the human ear canal, commercially available silicone ear-tips were attached to each speaker (Fig. 1). These tips introduced a measurable air gap between the grill and the ear-tip edge; this gap was quantified with a caliper and compensated for in the effective cavity length during analysis (Table I).

After assembly, the syringe length was adjusted to emulate various enclosure dimensions. A marker line on the syringe barrel was used as the insertion reference, and a ruler provided precise plunger positioning (Fig. 1). For example, to achieve an effective cavity length of 9 mm with speaker D, the plunger was set 7 mm from the marker to offset the 2 mm tip gap.

B. Data collection

The impedance of the earphones was measured using the Digilent Analog Discovery 2 device,[21] operated through the WaveForms software[22] and interfaced with our custom data acquisition system. Discrete single-tone stimuli spanning the full audible frequency range (20 Hz to 20 kHz) were presented sequentially at 501 linearly spaced frequency points. A 100 Ω reference resistor was connected in series with the device under test, replacing the 200 mΩ parallel configuration used previously,[13] in order to improve voltage sensitivity and numerical stability in impedance estimation. This modification enhanced the precision of both magnitude and phase estimation across the frequency range.

To automate and standardize the data acquisition process, a custom graphical user interface (GUI) was developed in Python using the Tkinter library.[23] The GUI communicated directly with the Analog Discovery 2, managing the playing of tones, automatic file naming, and run repetitions

TABLE I. Specifications of tested earbud speakers. The "Approximated gap" column indicates the value used for cavity-length correction during measurement.

Speaker ID	Earbud model and DC impedance	Measured gap (mm)	Approximated gap (mm)
A	Airline X complimentary earbud, 32.59 Ω	2.83	3
B	Airline Y complimentary earbud, 31.18 Ω	2.28	2
C	Sennheiser CX80S wired earbud, 27.35 Ω	2.12	2
D	Unbranded earbud (Amazon SG), 33.18 Ω	1.98	2

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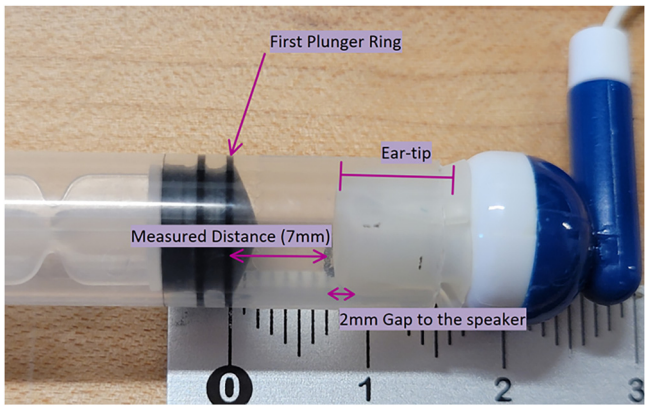


FIG. 1. Experimental enclosure setup using a syringe model to simulate ear-canal insertion depth.

(Fig. 2). Users selected the speaker ID, enclosure depth, and number of runs through drop-down menus, enabling consistent, repeatable data collection.

Measurements were acquired for four speakers across 14 enclosure lengths ranging from 5 mm to 29 mm at 3-mm intervals, supplemented by the additional key distances (9, 24, and 39 mm) tested in Kim *et al.*,[13] as well as "Open" and "Blocked" terminal conditions. Each configuration was repeated 100 times, yielding a comprehensive dataset of $4 \times 14 \times 100 = 5600$ impedance files for ML analysis.

For each sequentially applied single-frequency excitation, a CSV file was generated containing 501 rows of the following five parameters: (1) frequency, (2) impedance phase (degrees), (3) impedance magnitude, (4) real part (R), and (5) reactive component (X). The Python script then automatically organize the measurements for data parsing and analysis. Representative visualizations of the impedance magnitude, phase, and R–X relationships are presented in Sec. III.

C. Data preprocessing

Before ML training, the dataset was preprocessed in a structured manner. The program first identified the data directory according to the configuration table (config), which contained speaker labels and the set of enclosure length. For each CSV file, selected impedance fields (i.e., phase and magnitude) were extracted based on configuration parameters, then row indices were limited to the region of interest (within the 501 frequency points), resulting in a multi-dimensional feature tensor of shape (501, 2) for each data sample. The resulting tensor x was then min–max normalized to the range [0, 1], defined as follows:

$$x_{\text{norm}} = \frac{x - x_{\min}}{x_{\max} - x_{\min}}. \tag{1}$$

Depending on the configuration, either "the enclosure length alone" classification or the "combined speaker & enclosure length pair" classification served as the class label, representing the categorical value to be predicted by the DNN model. data. The data were divided into 60% training, 20% validation, and 20% testing sets, and output categories were one-hot encoded to enable multi-class learning. For a given sample with the true class label y, where $i \in \{1, 2, 3, \ldots, n\}$ denotes the set of possible class labels, one-hot encoding represents the class label as a vector $\mathbf{c}$ such that $c_y = 1$ and $c_i = 0$ for all $i \neq y$.[24] This representation provides a target indicator for the estimated class probability vector produced by the Softmax function as follows:

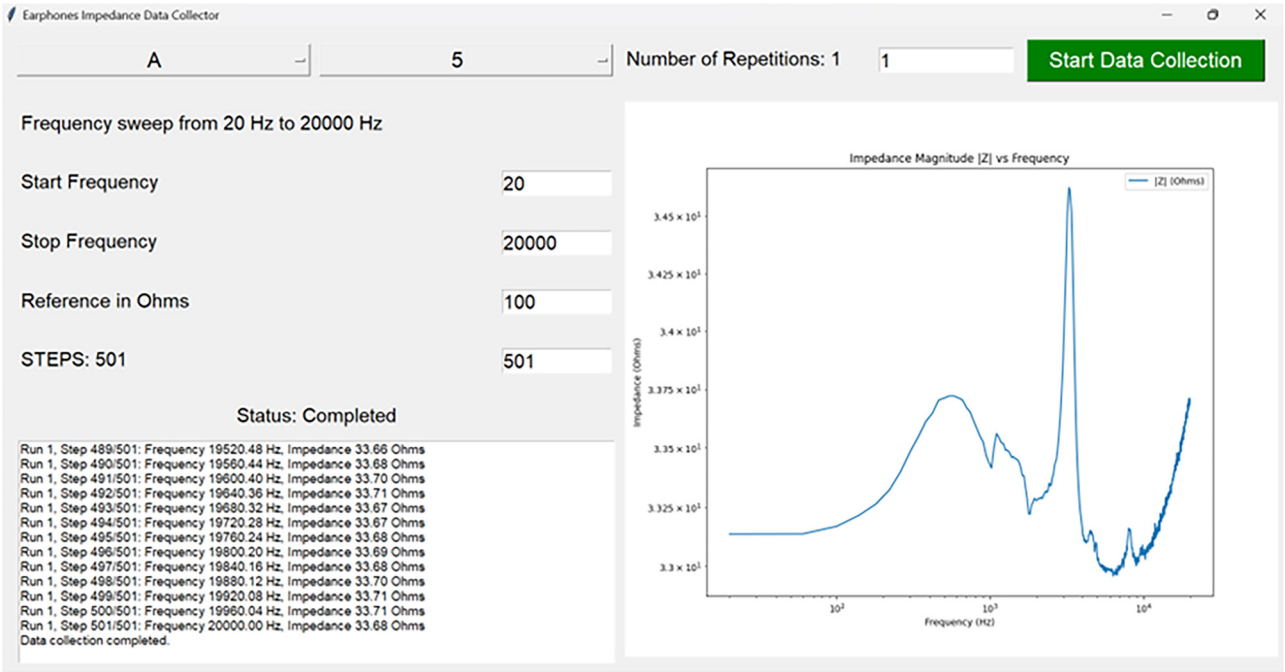


FIG. 2. Custom Python GUI for automated impedance data collection. Example shown: one run of speaker A at 5 mm enclosure depth.

$$\text{softmax}(x)_i = \frac{\exp(x_i)}{\sum_{j=1}^{n} \exp(x_j)}, \tag{2}$$

where $x = (x_1, \ldots, x_n)$ denotes the vector of pre-activation outputs of the output layer, and x_i corresponds to the output associated with the ith class.[24]

D. Model training and evaluation

We formulate the enclosure length prediction task as a multi-class classification problem and train the classification model to correctly predict the true class label y for each sample. Target class labels consisted of 14 enclosure lengths: Blocked, 5, 8, 9, 11, 14, 17, 20, 23, 24, 26, 29, 39, Open—across all four speakers. The goal of the classification model is to learn a complex representation of the underlying physical properties, such as phase and magnitude across all 501 frequencies, and to estimate the corresponding class label probabilities.

A supervised DNN was developed using the Keras Sequential API, a Python-based deep learning framework for defining and training neural network models.[25] The architecture, detailed in Algorithm 1, consists of four fully connected (Dense) layers with interleaved dropout for regularization.

Note that the network input is represented as a multi-dimensional feature tensor rather than a pre-flattened vector, enabling feature-wise processing in the early layers. The first three Dense layers act independently on each feature channel, and the resulting representations are concatenated via a flattening layer before being jointly processed by the final Dense layer. The ReLU activation function, defined as $\text{ReLU}(x) = \max(0, x)$, was applied to the outputs of all hidden layers to introduce nonlinearity into the network and to enable faster training compared to other activation functions. To produce normalized class likelihood estimates at the output layer, we apply the Softmax activation function [Eq. (2)].

Training employed the Adam optimizer with default parameters, an efficient stochastic optimization method that adaptively adjusts gradient updates while requiring little memory overhead.[26] For the loss function, the categorical cross-entropy loss $\mathcal{L}_{\text{CCE}} = -\sum_{i=1}^{n} y_i \log(\hat{y}_i)$, a standard choice for multi-class classification, was used to quantify the dissimilarity between the predicted class probability distribution ($\hat{y}_i$) and the one-hot encoded ground-truth labels (y_i). A batch size of 32 and an initial epoch of 200 were used to observe convergence and detect overfitting trends. The optimal epoch count was subsequently refined, effectively implementing a manual early-stopping criterion.

ALGORITHM 1. Pseudocode of the DNN architecture for impedance-based classification.

Input: input tensor x for each sample, num_classes
Output: trained_model;
Initialize a sequential model.
Add layers:
    Dense (128, ReLU, x);
    Dropout (0.2);
    Dense (256, ReLU);
    Dropout (0.5);
    Dense (128, ReLU);
    Dropout (0.2);
    Flatten ();
    Dense (num_classes, Softmax).
Return model.

Performance was evaluated using standard metrics (accuracy, precision, and recall) defined as follows:

$$\text{Accuracy} = \frac{\text{TP} + \text{TN}}{\text{Total}},$$
$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}}, \qquad \text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}, \tag{3}$$

where TP, TN, FP, and FN refer to true-positive, true-negative, false-positive, and false-negative classification results, respectively. Confusion matrices were also generated to visualize class-specific outcomes. Comparative results between pooled and speaker-specific training are summarized in Sec. III.

E. Regression model

Although enclosure length prediction was initially formulated as a classification task due to the discrete set of available enclosure lengths, we additionally investigated a regression formulation to assess whether explicitly modeling enclosure length as a continuous variable could improve predictive accuracy and generalization. From a physical standpoint, enclosure length is inherently continuous, and a regression model may in principle capture fine-grained relationships between impedance features and length that are not directly optimized in a categorical framework. Exploring a regression approach therefore serves as an important validation of whether the discretization imposed by classification limits performance, and whether a continuous prediction model can better exploit the underlying structure of the data. This comparison also allows us to assess the robustness of the classification-based approach by benchmarking it against a model that directly minimizes distance-based errors on the target variable.

The regression model was implemented by modifying the classification architecture to produce a single linear output corresponding to enclosure length, while keeping the remainder of the network unchanged as in Algorithm 2. Importantly, the same input features, preprocessing steps, data splits (training, validation, and test sets), optimizer settings, batch size, and training protocol were used for both the classification and regression models to ensure a fair comparison.

The classification model was trained using categorical cross-entropy loss, while the regression model was trained using mean squared error (MSE) loss. For the regression setting, enclosure lengths were treated as continuous values

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ALGORITHM 2. Pseudocode of the DNN architecture for impedance-based regression.

Input: input tensor x for each sample, num_classes
Output: trained_model
Initialize a sequential model.
Add layers:
    Dense (128, ReLU, input_shape of x);
    Dropout (0.2);
    Dense (256, ReLU);
    Dropout (0.5);
    Dense (128, ReLU);
    Dropout (0.2);
    Flatten ();
    Dense (1, linear).
Return model.

and scaled during training, with inverse scaling applied during evaluation. Data corresponding to the Open and Blocked conditions were excluded from regression experiments, as these conditions do not represent meaningful continuous enclosure lengths. Model performance was evaluated using standard regression metrics, including mean absolute error (MAE), root mean squared error (RMSE), and coefficient of determination (R^2). To enable a direct quantitative comparison with the classification model, classification outputs were converted to continuous enclosure length estimates using an expected-value formulation, computed as the probability-weighted sum of class-specific enclosure lengths as follows:

$$\hat{y} = \sum_{k=1}^{K} p_k \, y_k. \tag{4}$$

Here, K denotes the total number of discrete enclosure length classes, p_k is the predicted probability for class k obtained from the softmax output, and y_k is the enclosure length value corresponding to class k. These continuous estimates were then evaluated using the same MAE, RMSE, and R^2 metrics as the regression model, ensuring that both approaches were assessed under identical error measures.

III. RESULTS

In this section, we first present a qualitative visualization of the frequency-dependent impedance characteristics, followed by a quantitative analysis based on length prediction results for each cavity length, comparing the classification and regression models.

The electrical impedance data collected for each speaker under different cavity lengths were averaged and are shown in Figs. 3(a) and 3(b), illustrating the frequency-dependent magnitude and phase responses, respectively. Each trace was normalized to the value at the lowest measurement frequency (20 Hz) to facilitate relative comparison and mitigate systematic deviations due to instrumentation or environmental drift. This normalization highlights frequency-dependent impedance variations rather than absolute magnitude differences, allowing key resonance trends and anomalies to be visually distinguished. The data at 20 Hz exhibited a 3%–5% variation across 100 repeated measurements for each condition, confirming strong repeatability. The following subsections present the AI-based length prediction results both with and without speaker-specific differentiation.

A. Length prediction with speaker-specific differentiation

When the dataset included both speaker identity and enclosure length in the output label, the classification model achieved a test accuracy of 91.3%, with a precision of 93.2% and recall of 91.3% after 110 epochs. These results indicate that explicitly accounting for speaker-specific characteristics improves classification performance when modeling variability across different electroacoustic profiles within the evaluated dataset. Evaluation metrics are summarized in Table II, and the corresponding confusion matrix is shown in Fig. 4(a) where the colorbar denotes the number of samples in each confusion-matrix bin.

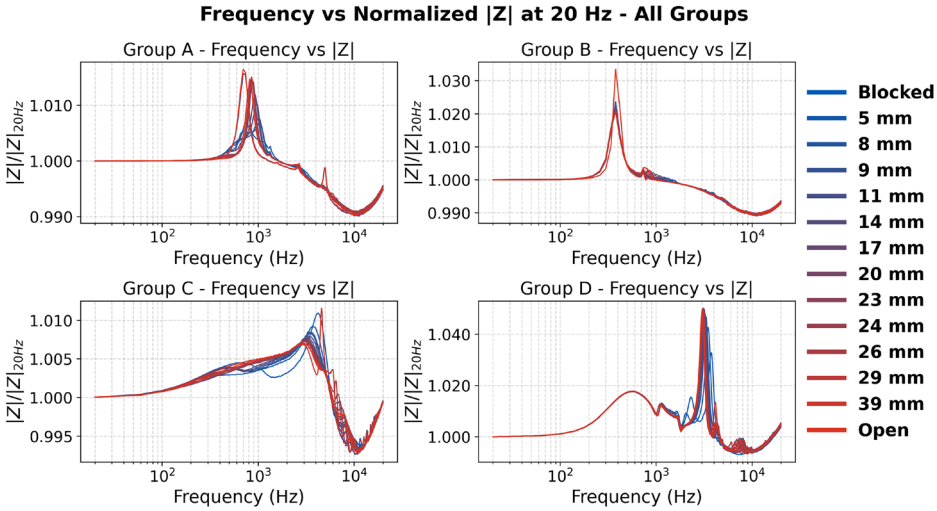
B. Length prediction independent of speaker type

When speaker labels were excluded and only enclosure length was used as the output label, the classification model achieved slightly lower performance: a test accuracy of 87.1%, precision of 89.6%, and recall of 86.9% after 100 epochs. The metrics are listed in Table II, and the confusion matrix is shown in Fig. 4(b). Although this configuration demonstrated reduced accuracy, it confirmed the robustness of the impedance–length relationship even without prior knowledge of the speaker's impedance profile.

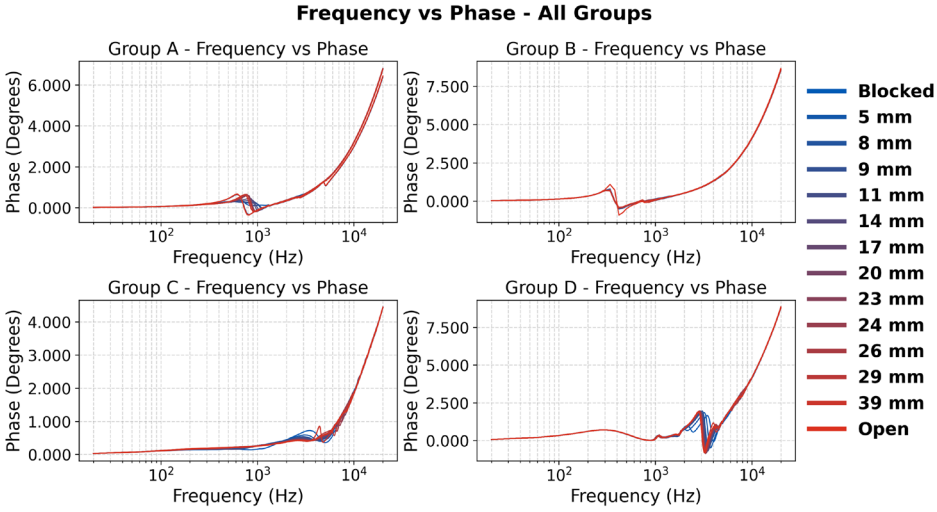
C. Length prediction for individual speakers

To investigate inter-speaker variability, models were trained separately on each speaker dataset. As summarized in Table III, the classification model for speaker C and speaker D achieved near-perfect results (accuracy $\approx$ 99.6%), significantly outperforming the combined datasets. Speaker A reached a comparable accuracy to the differentiated full model, whereas speaker B exhibited the lowest accuracy (58.6%), indicating higher internal variability or inconsistent impedance–length correlation. The individual confusion matrices are shown in Fig. 5.

To further substantiate the experimental findings, we examine the impact of frequency-domain feature selection on model performance, demonstrating that restricting the analysis to acoustically informative frequency regions improves classification accuracy. Because speaker B yielded the lowest overall performance, further analysis was conducted by constraining training to specific frequency bands. The goal was to identify spectral regions that contributed most to predictive accuracy by visually examining the averaged magnitude and phase responses to identify frequency regions containing resonance features or impedance variations that changed systematically with acoustic loading. As



(a) Frequency vs. normalized impedance magnitude



(b) Frequency vs. impedance phase

FIG. 3. Averaged impedance characteristics of all speakers across test conditions.

shown in Fig. 3(a), speaker B exhibited distinct resonant variations across frequency bands.

Training restricted to the 619.4–2018 Hz range yielded the highest accuracy (89.6%) with 80 epochs—an improvement from 58.6% to 89.6% compared to the full-range dataset (Table IV). This result demonstrates that mid-frequency bands carry dominant discriminative features for impedance-based depth estimation.

Conversely, training on 20–539.48 Hz achieved comparable performance to the full range (60.4% accuracy), while the high-frequency range (2.057–20 kHz) performed poorly (7.1% accuracy). The corresponding confusion matrices are shown in Fig. 6.

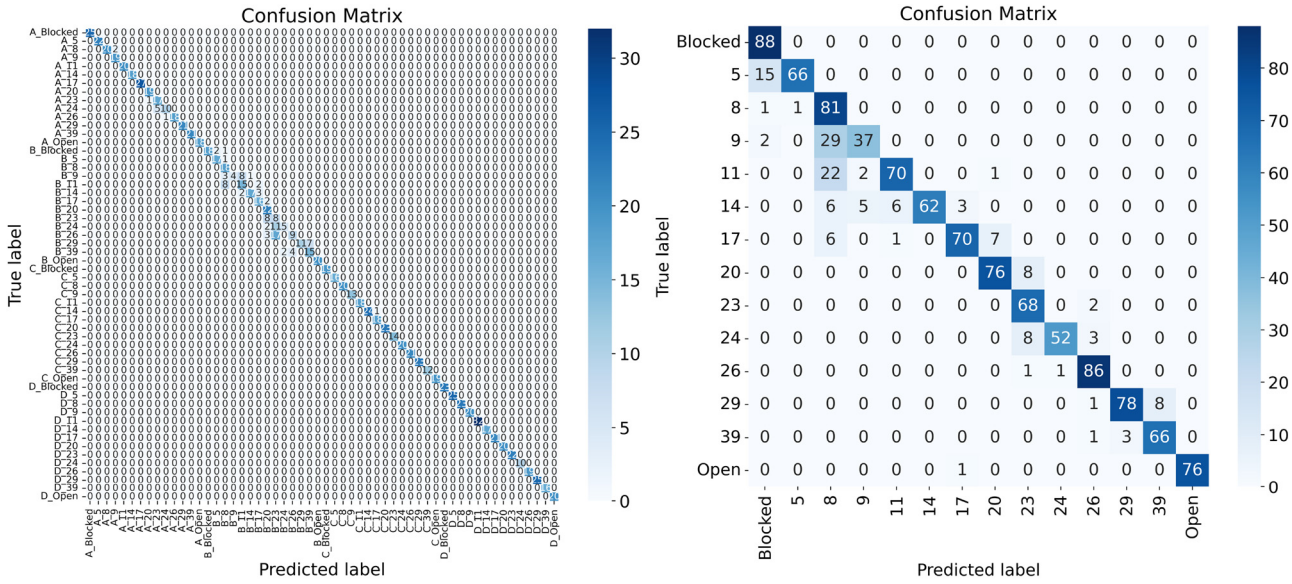
To examine whether the frequency selection strategy generalizes across speakers, the same frequency-cropping procedure was applied to speakers A, C, and D. Results confirmed that optimal spectral ranges vary by transducer but can further refine accuracy.

For speaker A, restricting the range to 20 Hz to 10 kHz improved accuracy to 98.2% after 125 epochs. For speaker C, the same range achieved 99.6% accuracy after 100 epochs—maintaining performance while requiring more training cycles. For speaker D, focusing on the 1–10 kHz range resulted in 99.3% accuracy, comparable to its full-range result. Notably, frequency content above 10 kHz did not contribute to improvements in classification performance across all speakers, suggesting that impedance data collection can be effectively limited to frequencies below 10 kHz without loss of predictive accuracy. It is also worth noting that the earphone specifications do not support such

TABLE II. Performance comparison for models trained with and without speaker differentiation.

	Speaker and length	Only length
Epoch	110	100
Accuracy	91.3%	87.1%
Precision	93.1%	89.6%
Recall	91.3%	86.9%

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(a) Trained with speaker-specific differentiation

(b) Trained without speaker differentiation (length-only labels)

FIG. 4. Confusion matrix for length prediction model trained with and without speaker-specific differentiation. The colorbar presents the number of samples in each confusion-matrix bin.

high frequencies. Collectively, these findings demonstrate that frequency-based feature selection can enhance efficiency and performance for certain transducer types while confirming the high reproducibility of the proposed impedance-based sensing approach.

D. Classification vs regression

The classification model and regression model were trained and evaluated on a dataset with all four speakers and output of pure enclosure length, excluding speaker labels. Table V summarizes the performance of the classification and regression formulations evaluated using continuous regression metrics. When classification outputs were converted to continuous length estimates using the expected-value formulation, the classification model achieved slightly lower MAE and RMSE and a marginally higher R^2 compared to the direct regression model. However, the overall differences between the two approaches were small, indicating that regression performance was only marginally reduced relative to classification.

While discretizing enclosure length into a finite number of classes could, in principle, yield optimistic performance relative to a fully continuous prediction task, our results indicate robust algorithm performance under both formulations. Both approaches exhibit increasing prediction error at larger enclosure lengths, as shown in Figs. 7 and 8, indicating reduced accuracy in regions with fewer available training samples. This trend suggests that performance limitations at the upper length range are driven more by data distribution effects than by the modeling formulation itself.

TABLE III. Performance metrics for models trained on individual speaker data.

	A	B	C	D
Epoch	125	110	50	50
Accuracy	92.9%	58.6%	99.6%	99.6%
Precision	92.9%	62.7%	99.7%	99.7%
Recall	93.3%	59.3%	99.6%	99.5%

IV. DISCUSSION

The results of this study should be interpreted through the lens of electroacoustic coupling, where changes in acoustic loading manifest as nonlinear perturbations in the speaker's electrical impedance. The nonlinear behavior observed in this study—manifested as resonance frequency and impedance peak shifts with changing acoustic load—reflects the intricate coupling between electrical, mechanical, and acoustic domains. These inter-domain dynamics are difficult to capture through purely analytical or lumped-parameter models, motivating the adoption of data-driven approaches.

The DNN effectively captured the nonlinear dependencies between acoustic volume, compliance, and resonance frequency. Unlike conventional linear regression or physics-based models, DNNs employ nonlinear activation functions (e.g., ReLU) and gradient-based optimization to discover latent relationships that may not be explicitly modeled. Prior studies have emphasized the value of such data-driven models for capturing nonlinear transducer dynamics and predicting electroacoustic behavior with high fidelity.[19,27,28] In the context of this work, the role of ML is not to replace physical understanding, but to complement it by learning mappings that arise from complex boundary-condition

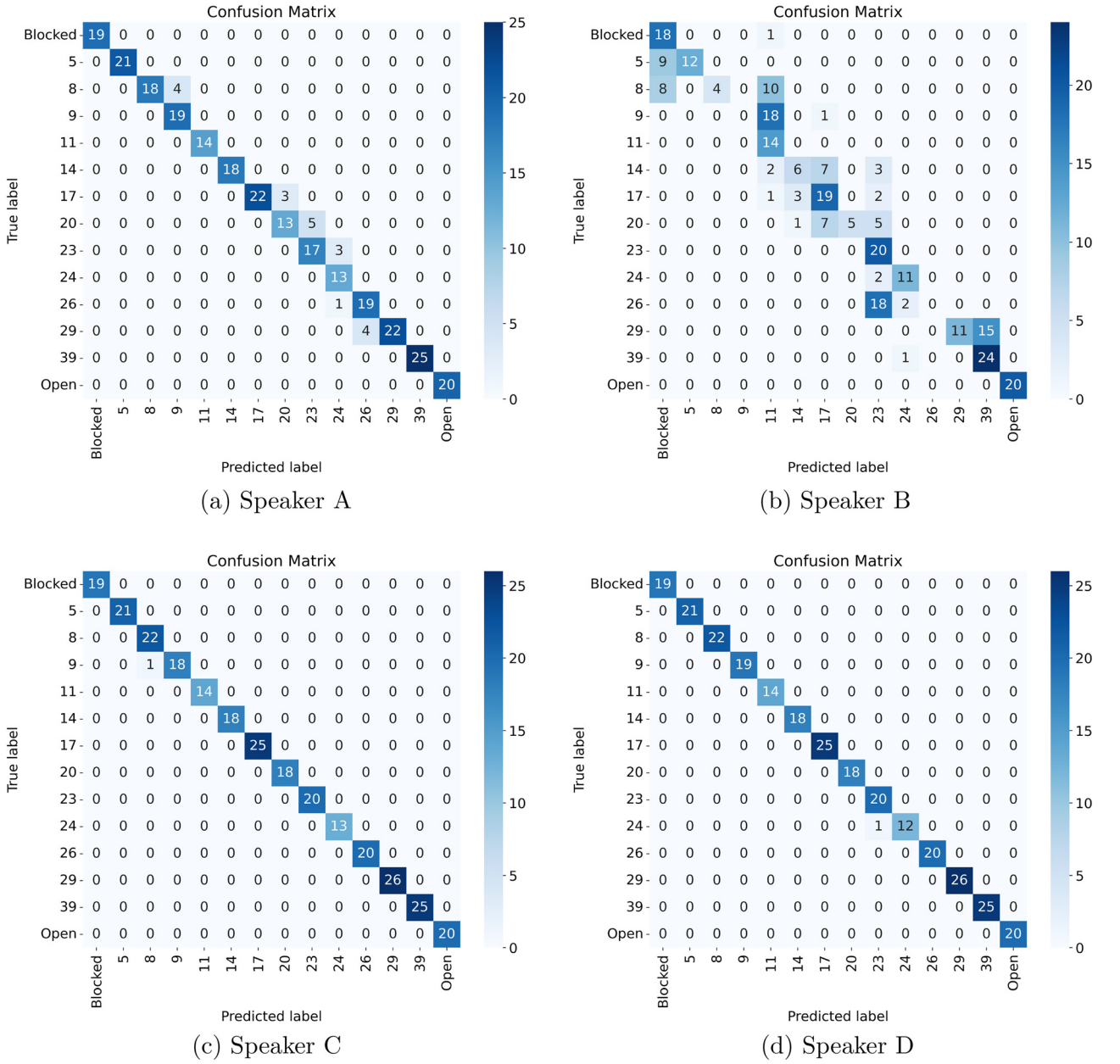


FIG. 5. Confusion matrices for length prediction trained on individual speaker datasets. The colorbar of each matrix indicates the number of samples in each matrix bin.

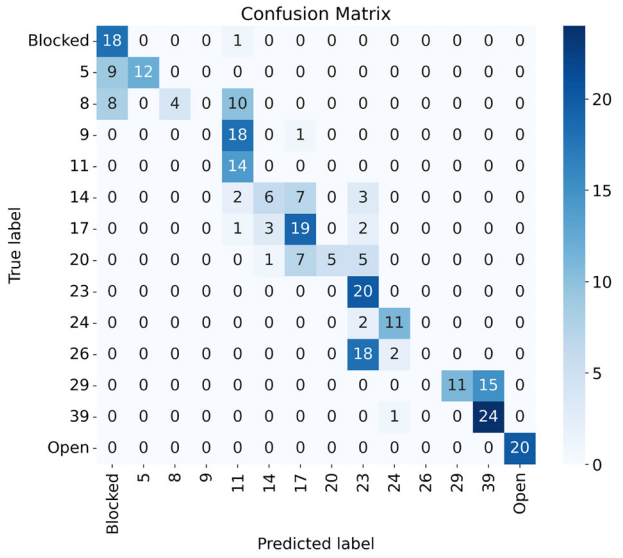
variations. The present results reinforce that capability: high prediction accuracy across multiple speaker types indicates that impedance patterns encode sufficient information to infer insertion depth, even in the presence of inter-device variability.

TABLE IV. Comparison of model performance for speaker B under each frequency cropping range.

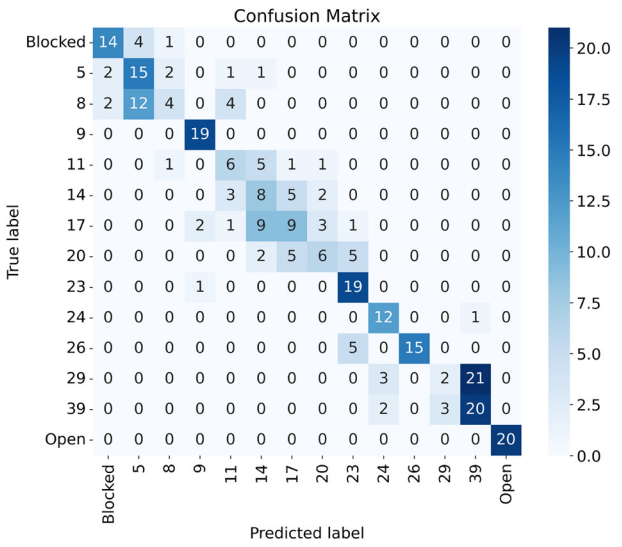
	All	20–539.48 Hz	619.4–2018 Hz	2000–20 000 Hz
Epoch	110	95	80	7
Accuracy	58.6%	60.4%	89.6%	07.1%
Precision	62.7%	60.8%	91.7%	0.5%
Recall	59.3%	62.1%	91.6	7.1%

All impedance measurements were conducted within 20 Hz–20 kHz, the human auditory range, ensuring practical relevance for hearing-device applications. Operating in this frequency range also ensures that the observed impedance variations are directly tied to acoustically meaningful resonances rather than ultrasonic or parasitic effects. To improve classification performance and reduce the influence of weakly informative spectral components, we performed frequency-domain diagnostics to identify acoustically sensitive frequency regions and restricted the classification model input accordingly. This procedure represents a physically motivated feature selection strategy rather than a direct interpretation of the internal neural network representations.

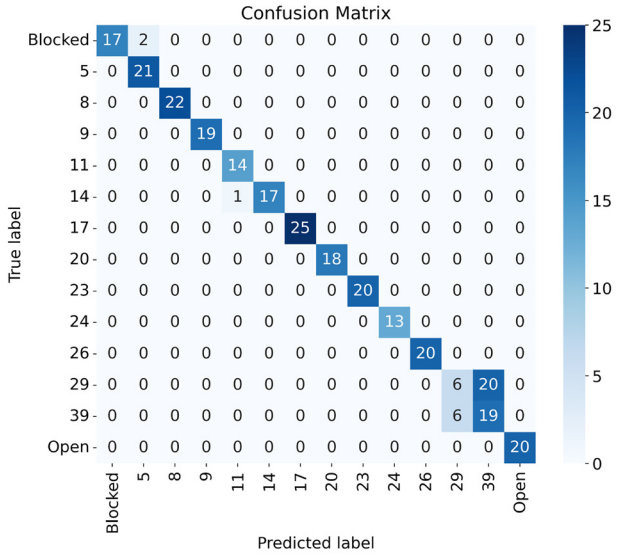
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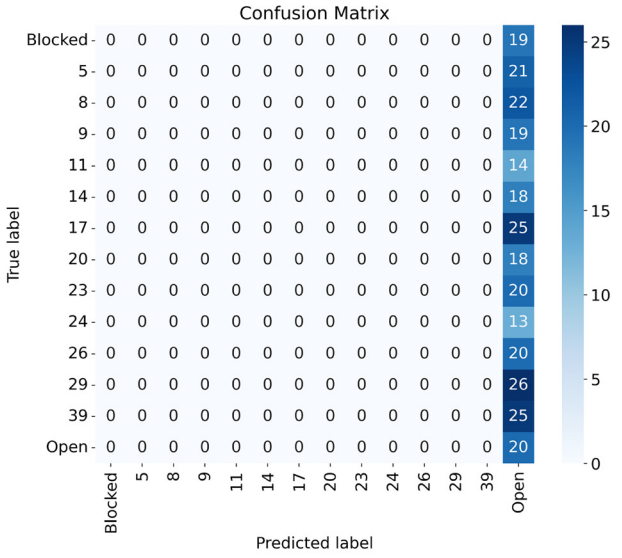
(a) Full range (20–20 kHz): duplicated



(b) Low range (20–539.48 Hz)



(c) Mid range (619.4–2018 Hz)



(d) High range (2.057–20 kHz)

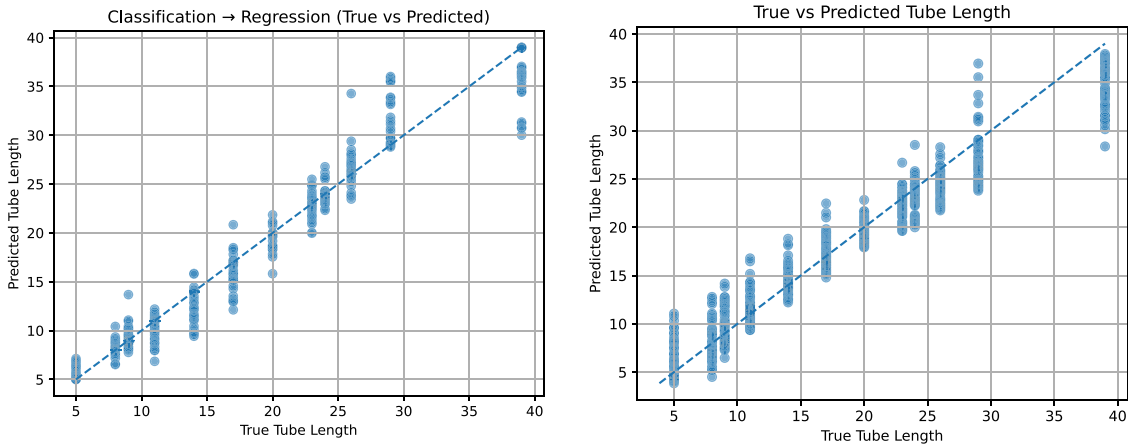
FIG. 6. Confusion matrices for speaker B models trained on cropped frequency ranges.

Speaker B provided a particularly illustrative case. Its baseline model exhibited poor performance (accuracy = 0.586; Table III), prompting a detailed examination of its magnitude and phase spectra (Fig. 3). Distinct spectral differentiability was observed in the mid-frequency range. These findings are consistent with the expectation that mid-frequency resonances are especially sensitive to acoustic compliance and cavity length. Guided by this finding, frequency-cropping experiments isolated ranges yielding optimal discrimination (Table IV), improving accuracy to 0.896 within 619.4–2018 Hz. This substantial gain validates that frequency-selective data preprocessing can enhance both model accuracy and robustness by mitigating frequency-domain noise or irrelevant information.

Moreover, narrowing the frequency range reduced the input dimensionality and, consequently, the total model size by approximately 30%, improving computational efficiency without loss of precision. This result highlights an important

TABLE V. Performance comparison between classification-based and regression-based enclosure length estimation.

	Classification	Regression
Epoch	110	150
MAE	0.5381	1.7294
RMSE	1.3125	2.3054
R^2	0.9812	0.9421



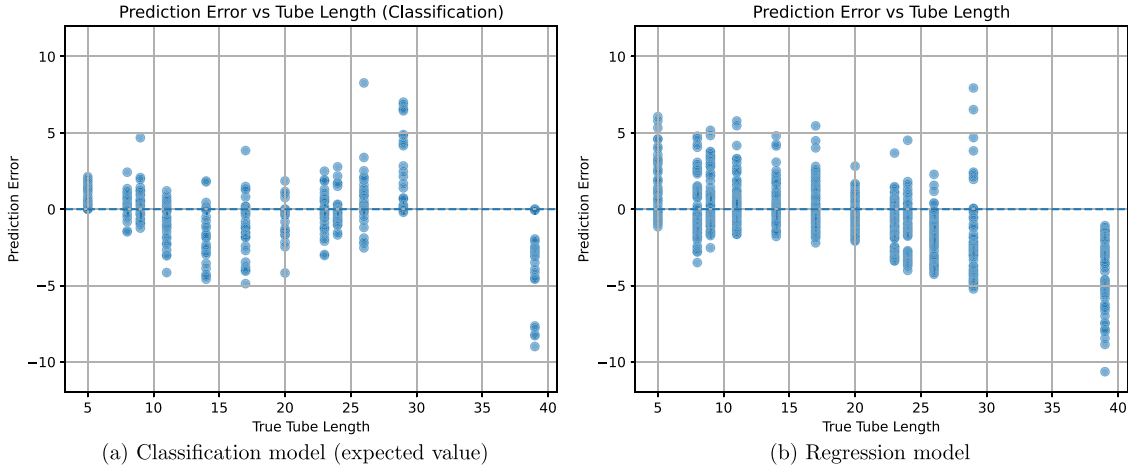
(a) Classification model (expected value) (b) Regression model

FIG. 7. True versus predicted enclosure length for classification-based (a) and regression-based (b) models.

practical implication: physics-informed spectral selection can simultaneously improve interpretability, efficiency, and predictive performance. It can be interpreted through both acoustic and modeling considerations as well. At very low frequencies, earphone transducers typically exhibit reduced electromechanical sensitivity, and the dimensions of the ear canal are much smaller than the corresponding acoustic wavelengths. Under these conditions, the acoustic pressure distribution within the cavity is nearly uniform, and changes in cavity length or compliance produce relatively weak perturbations in the electrical impedance response. As a result, low-frequency components tend to carry limited discriminative information with respect to acoustic loading in the present configuration. In contrast, mid-frequency regions are more strongly influenced by coupled electroacoustic resonances, where changes in boundary conditions lead to measurable and systematic variations in impedance magnitude and phase.

From an AI modeling perspective, the DNN employed in this study treats all input frequency components as potential features and does not inherently distinguish between physically informative and weakly informative regions. When the full frequency range is used as input, especially for speakers exhibiting relatively flat or noisy spectral responses, the presence of low-sensitivity frequency components can dilute the contribution of acoustically relevant features and hinder effective learning. By restricting the input to frequency regions that exhibit the strongest and most consistent sensitivity to acoustic loading, the feature dimensionality is reduced and the relative contribution of physically meaningful impedance variations is enhanced. This combination of acoustically informed frequency selection and data-driven inference helps explain the pronounced performance improvement observed for the initially poorly performing speaker, and underscores the benefit of integrating physical insight with neural network–based modeling.

The classification-based approach achieved lower numerical error than the direct regression model when evaluated on the discrete test set, even after converting classification outputs to continuous enclosure length estimates using an expected-value formulation. This outcome is reasonable given the discrete nature of the dataset and the training objective of the classification model. Specifically, the classification network is optimized using a cross-entropy



(a) Classification model (expected value) (b) Regression model

FIG. 8. Prediction error versus true enclosure length for classification-based (a) and regression-based (b) models.

loss, which encourages probability mass to concentrate on the correct discrete enclosure length class. When the predicted probability distribution is sharply peaked, the resulting expected-value estimate closely matches the true enclosure length. Although this expected-value output is continuous, it remains constrained to combinations of predefined class centers, providing an implicit regularization that aligns well with the discretized evaluation targets.

The regression model exhibits higher error on the discrete evaluation set, but its prediction errors remain within an acceptable range for practical use. Importantly, the regression approach offers improved flexibility and is better suited for applications involving interpolation or previously unseen enclosure lengths, highlighting a trade-off between numerical performance on discretized data and generalization to continuous operating conditions. For instance, as shown in the dataset, shorter enclosure lengths are more densely represented, while longer lengths are comparatively sparse. This imbalance biases the learning process toward regions of the output space with higher sample density, resulting in reduced accuracy for underrepresented length values. In the regression setting, this effect is particularly pronounced, as the regression model is required to learn a continuous mapping across the entire length range without explicit class boundaries. Still, its prediction errors remain within an acceptable range for practical use. Importantly, the regression approach offers improved flexibility and is better suited for applications involving interpolation or previously unseen enclosure lengths, highlighting a trade-off between numerical performance on discretized data and generalization to continuous operating conditions.

Addressing limitation regarding the imbalanced dataset would require additional measures, such as rebalancing the training data, applying sample-weighted loss functions, or collecting additional measurements at longer enclosure lengths. These approaches are left for future work and are expected to further improve regression performance, particularly in sparsely sampled regions of the length domain.

V. CONCLUSION

This study demonstrates the feasibility of repurposing speakers (earphones)—traditionally audio output devices—as self-sensing transducers through electroacoustic coupling (motional impedance). By integrating impedance measurements with AI-based analysis, the system accurately inferred acoustic loading conditions and insertion depth across multiple speaker types. Overall, impedance-derived features paired with a compact DNN classification yielded high-accuracy length prediction across transducers. Including speaker identity improved performance (accuracy = 91.3%, precision = 93.1%, recall = 91.3%), while a length-only model still performed robustly (accuracy = 87.1%, precision = 89.6%, recall = 86.9%), confirming a generalizable impedance–length mapping. Per-speaker training revealed near-perfect results for speakers C and D (accuracy ≈ 99.6%), parity for speaker A (accuracy = 92.9%), and a challenging case for speaker B (accuracy = 58.6%). Physics-guided frequency cropping markedly improved speaker B (619.4–2018 Hz; accuracy = 89.6%; 31% percentage points vs full range) and maintained or enhanced accuracy for other speakers while reducing bandwidth and training time. Normalization at 20 Hz and averaging over 100 runs produced repeatable spectra (3%–5% variation), underpinning model stability.

The results highlight that measurable impedance variations encode physical information about the surrounding acoustic environment, validating the potential of earphones as low-cost, self-referenced sensors. Compared to the authors' previous image-based pipeline,[13] the present raw-impedance framework is simpler, more data-efficient, and offers interpretable, frequency-localized performance gains. Additionally, targeted frequency-domain analysis improved performance and model transparency. The findings collectively establish a foundation for developing adaptive, impedance-driven diagnostics for ear health assessment and extend the broader applicability of electroacoustic transducers to biomedical and structural monitoring domains. In future work, we plan to extend the proposed electrical impedance sensing technique to human-ear–like ear canal models, such as binaural head simulators or physical ear canal replicas with known geometry and controlled variations in canal depth and boundary conditions. The regression model represents a practical and realistic approach for real-time test. Future work will compare pooled and speaker-dependent regression strategies to evaluate the effect of inter-speaker variability. In addition, because electrical impedance is sensitive to the surrounding acoustic environment, this framework may be extended to sense acoustic parameters beyond insertion depth, including changes in cavity compliance and boundary conditions. Also, refining the selected frequency range (particularly at low frequencies) and exploring adaptive methods for selecting optimal frequency bands will be an important direction for future work.

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AUTHOR DECLARATIONS

Conflict of Interest

The authors declare no conflicts of interest. All coauthors have reviewed and approved the contents of the manuscript, and there are no financial interests to disclose. We confirm that this submission is original work and is not currently under review by any other publication.

DATA AVAILABILITY

Data are available upon reasonable request from the corresponding author.

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